

SHAOHANG WEI

Github: github.com/sylvain-wei ◇ Wechat: [sylvain4ai](#)
(+86) 18983042867 ◇ shaohang@stu.pku.edu.cn

EDUCATION

Peking University

Ph.D. student in Computer Science and Technology

Sept. 2024 - Present

Supervisor: *Prof. Houfeng Wang*

- Research Interest: *Post-training, LLM/Agentic Reasoning*
- Affiliation: *Institute of Computational Linguistics, School of Computer Science, Peking University*

Beihang University

Bachelor of Engineering in Artificial Intelligence

Sept. 2020 - Jun. 2024

EXPERIENCE

Qwen Pilot, Tongyi Lab, Alibaba Cloud Computing Co. Ltd

May. 2025 - Dec. 2025

Research Intern

RL for LLM , Post-training , Reasoning

- Token-Level Distributional Analysis:
 - Conducting a granular analysis of RLVR (GRPO/DAPO) training dynamics, revealing that RL induces sparse but critical distributional shifts (only $< 5\%$ tokens diverge significantly).
 - Proved via cross-sampling experiments that injecting a small fraction of RL-selected tokens into base generations recovers most performance gains, providing the theoretical foundation for sparse teacher-guidance strategies.
- Efficient Exploration via Speculative Teacher-Guidance
 - *Algorithm*: Proposed a hybrid exploration strategy where a converged Teacher Model guides the Policy Model during rollouts by resampling tokens at positions with high entropy or divergence. Achieved 40.9% (Pass@1:32) on AIME24, surpassing the DAPO baseline (37.4%).
 - *Engineering Challenge*: Naive implementation of teacher-guidance required synchronized forward passes at every step, increasing rollout latency from 3 mins to 40 hours per prompt.
 - *Solution (Speculative Decoding Adaptation)*: Leveraged the theoretical finding of "sparse divergence" (2%-8% replacement rate) to implement a Speculative-Decoding-based rollout mechanism. The Policy Model generates draft chunks (e.g., 10 tokens), and the Teacher Model verifies/intervenes only when necessary.
 - *Result*: Reduced training overhead by 15x, restoring rollout latency to near-standard levels while maintaining the algorithmic performance gains.

AISphere (Pixverse.ai)

May. 2024 - Aug. 2024

Algorithm Intern

- Synthesized high-quality data for PixVerse by using closed-source and open-source models for video captioning. Analyzed data quality with NLTK and SpaCy, and performed text extraction from video subjects using GPT-4o and a large-scale visual language model. Improved the SBS annotation framework, launched the evaluation platform, and optimized video frame rates to balance caption quality and length.

Beihang University

Undergraduate Research Assistant (Supervisor: Fuzhen Zhuang)

May. 2023 - Jul. 2024

- Built a human-centric evaluation framework (MindScore) to address the lack of reliable methods for assessing text-to-image foundation models. Designed a four-module scoring system capturing semantic alignment and psychological image quality, and validated it on three benchmark datasets with human preference annotations. Demonstrated clear performance gains over state-of-the-art baselines and provided new insights into T2I model behavior.

Beihang University

Apr. 2023 - Sept. 2023

Undergraduate Research Assistant (Supervisor: Jianxin Li and Qingyun Sun)

- Developed a method for modeling higher-order relationships in graphs using a GCN-based attention mechanism. Introduced multi-view contrastive learning for self-supervised node representation, improving performance. Outperformed the SOTA model GRACE on citation datasets (Cora, CiteSeer, PubMed, DBLP).

PUBLICATIONS

1. **S. Wei**, W. Li, F. Song, W. Luo, T. Zhuang, H. Tan, Z. Guo, H. Wang
TimE: A Multi-level Benchmark for Temporal Reasoning of LLMs in Real-World Scenarios
NeurIPS 2025 (Spotlight, top 3%), December 2025.
General Domain Reasoning (Temporal) , RAG
Long-context , Memory
2. H. Meng, K. Huang, **S. Wei**, C. Ma, S. Yang, X. Wang, G. Wang, B. Ding, J. Zhou
Sparse but Critical: A Token-Level Analysis of Distributional Shifts in RLVR Fine-Tuning of LLMs
ICLR 2026, April 2026.
RLVR , Reasoning
3. W. Luo, F. Song, W. Li, G. Peng, **S. Wei**, H. Wang
Odysseus Navigates the Sirens' Song: Dynamic Focus Decoding for Factual and Diverse Open-Ended Text Generation
ACL 2025 Main, August 2025.
Hallucination , Decoding
4. F. Song, **S. Wei**, W. Luo, Y. Fan, T. Liu, G. Wang, H. Wang
Well Begun is Half Done: Low-resource Preference Alignment by Weak-to-Strong Decoding
ACL 2025 Findings, August 2025.
Weak-to-strong , Alignment , Decoding
5. Y. Zhu, **S. Wei**, X. Wang, K. Xue, X. Zhang, S. Zhang
MeNTi: Bridging Medical Calculator and LLM Agent with Nested Tool Calling
NAACL 2025 Main, April-May 2025.
Agent
6. Y. Tong†, J. Zhang†, **S. Wei†**, W. Guo, F. Zhuang, D. Wang, X. Yang, R. Xuan
MindScore: quantifying human preference for text-to-image generation through multi-view lens
SCIS 2025 (CCF-A).
Text-to-image , Multi-modal LLM , Contrastive Learning

†: Equal contribution.

SELECTED PREPRINTS

1. W. Li, X. Zhang, **S. Wei**, Y. Gao, Z. Guo, W. Luo, F. Song, Y. Huang, H. Wang
SWE-Ext: Extending and Scaling Augmented Data for Repository-Level Coding Tasks
Submitted to ACL 2026
SWE , Code

2. W. Luo, G. Peng, W. Li, **S. Wei**, F. Song, L. Wang, N. Yang, X. Zhang, J. Jin, F. Wei, H. Wang
Two Pathways to Truthfulness: On the Intrinsic Encoding of LLM Hallucinations
Submitted to ACL 2026
Hallucination , Interpretability
3. F. Song, **S. Wei**, B. Gao, Y. Wang, W. Luo, W. Li, L. Yao, W. Xiong, L. Chen, T. Liu, H. Wang
Mitigating Overthinking through Reasoning Shaping
Submitted to ACL 2026
Efficient Reasoning , RLVR
4. W. Yu, **S. Wei**, J. Liu, Y. Li, M. Hu, A. Liu, H. Zhang, I. King
Probability-Entropy Calibration: An Elastic Indicator for Adaptive Fine-tuning
Submitted to ICML 2026
Reasoning , Post-training
5. S. Yang, J. Lu, K. Huang, C. Ma, **S. Wei**, Y. Liu, G. Wang, J. Zhou, L. Yuan
One-Way Policy Optimization for Self-Evolving LLMs
Submitted to ICML 2026
RLVR , Reasoning
6. J. Lu, K. Huang, J. Wu, S. Yang, J. Li, C. Ma, **S. Wei**, X. Wang, G. Wang, J. Zhou
Experience Augmented Policy Optimization for LLM Reasoning
Submitted to ICML 2026
RLVR , Reasoning
7. Z. Xu, **S. Wei**, Z. Han, J. Jin, Z. Yang, X. Li, H. Tan, Z. Guo, H. Wang
CiteCheck: Towards Accurate Citation Faithfulness Detection
arXiv preprint arXiv:2502.10881
Fact-checking

AWARDS AND SCHOLARSHIPS

First Prize	November 2021
<i>The 13th National College Students Mathematics Competition (Non-Mathematics Category)</i>	
First Prize	November 2021
<i>The Beijing Mathematics Competition</i>	
Honor Prize	April 2023
<i>The American Mathematical Contest in Modeling (MCM)</i>	
Second Prize	November 2022
<i>The Beijing Division Group A of the 2022 Higher Education Press Cup National College Students Mathematical Modeling Competition.</i>	
Song Qingling Future Scholarship	May 2024
Excellent Graduate	May 2024
<i>Excellent Graduate of Beihang University</i>	
Excellence in Study Scholarship	September 2021, 2022, 2023, 2024
<i>Excellence in Study Scholarship of Beihang University</i>	
Excellence in Contest Scholarship	September 2022, 2023
<i>Excellence in Contest Scholarship of Beihang University</i>	

SKILLS

Programming Languages and Frameworks

C, C++, Python, Shell

VeRL, Llama-factory, HuggingFace, PyTorch

Languages

English (CET-6 619; CET-4 657); French (Beginner Level)